

DESIGN DOCUMENT

1. Title

Project Name: **EMULATING A SIMPLE 16-BIT MICROPROCESSOR.**

Authors: **GAUTHAM S THAMPI**

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2. Overview

The objective of this project is to design and implement a C++ program that emulates the core functionality of a simple 16-bit microprocessor. The simulation will be with respect to register operations, memory access, instruction parsing and arithmetic operations using object-oriented programming principles.

3. Specifications

- 16-bit Microprocessor
- Four General purpose registers (AX, BX, CX, DX)
- Memory of 64 KB, byte addressable memory

4. Proposed Solution

- Used Classes and functional units for creating the structure.
- Classes were defined for Registers, Memory and Instructions and datas are encapsulated.
- Instructions are parsed, validated and executed according to the specifications

Classes

- Register – Adding, Manipulating and Storing registers (AX, BX, CX, DX)
- Memory – Allocating and Managing Memory
- Instructions – To store, edit and load the instructions to be executed.

5. Conclusion

- This system properly manages the addition of values to the respective registers/memory allocation according to the input with respect to the allocated memory.